Pratapani Sai Nishant

ce23btech11041@iith.ac.in | nishant.pratapani@gmail.com | +91 9493439897 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.30
FIITJEE Junior College	Class XII (TSBIE)	2023	96.70%
FIITJEE SCHOOL	Class X (SSC)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

Selected as one of only **8 students** institute-wide for the Project Based Learning (PBL) programme in collaboration with **Ritsumeikan University, Japan**; collaborative transportation planning research with an on-site visit to Japan.

Awarded a full scholarship for academic excellence during Intermediate education.

PROFESSIONAL EXPERIENCE

AI Engineering Intern -- Civil Engineering Automation | KynosLogic Ltd. | Nagoya, Japan

- Built an end-to-end Python pipeline to extract horizontal and vertical road alignment data from DXF/CAD drawings and generate valid LandXML 1.2 output; integrated the Claude API for generalized vertical profile extraction using autonomous Y-coordinate clustering, achieving 100% accuracy against rule-based ground truth across all test files.
- Engineered recursive DXF block traversal to resolve nested INSERT entity access across 100+ layers and implemented median-based coordinate translation with automated outlier filtering to resolve multi-file coordinate system mismatch between plan view drawings.

Summer Intern -- Transportation Engineering | CDM Smith Inc. | Doha, Qatar

- Analyzed 7-day Automatic Traffic Counts, Turning Movement Counts, and Origin-Destination survey data across multiple municipalities; converted volumes into Passenger Car Units and developed trip distribution matrices for corridor-level planning.
- Conducted junction capacity analysis for 2 intersections using SIDRA Intersection software; evaluated 2030 traffic scenarios at 6% annual growth, identified operational deficiencies, and recommended geometric and signal timing improvements.
- Designed 16 bus stops and 1 km of Vehicle Restraint System, and quantified 361,901 m^2 of pavement layers using AutoCAD for detailed infrastructure planning.

PROJECTS

Driver Behavior Classification at Highway Entry Ramp Terminals using Deep Sequential Models

- Developed a trajectory-based behavior classification framework using ADAS vehicle trajectory data from Hyderabad ORR entry ramps, involving preprocessing, rolling-window feature extraction, and temporal signal smoothing.
- Generated latent driving behavior classes using unsupervised clustering over velocity and longitudinal acceleration statistics, identifying heterogeneous merging patterns across 155+ vehicle interactions.
- Implemented CNN-LSTM and CNN-BiLSTM architectures for sequential behavior learning, achieving 98.5% classification accuracy and demonstrating strong temporal representation capability for merge-zone behavior prediction.

Bayesian Physics-Informed Neural Networks for Inverse Estimation of Elastic Modulus in Concrete

- Developed a Bayesian Physics-Informed Neural Network (B-PINN) framework for inverse estimation of elastic modulus and Poisson's ratio in concrete using embedded PZT sensor measurements.
- Integrated governing wave-equation constraints with voltage-response data to infer material parameters while quantifying epistemic uncertainty through stochastic variational inference.
- Implemented physics-constrained optimization and automatic differentiation, incorporating Bayesian posterior estimation and uncertainty-aware parameter learning.

Detecting Deceptive Behavior in Vision-Language Models for Geospatial Interpretation *(Ongoing)*

- Investigating whether small vision-language models (VLMs) genuinely ground outputs in pixel-level geospatial data (NASA GPM IMERG precipitation imagery) or default to manipulated textual annotations such as titles, labels, colorbars, and filenames.
- Designed a pixel-invariant experimental framework across real extreme-precipitation events and built an evaluation pipeline (open VLMs, automated text-similarity metrics, LLM-as-judge scoring) to develop a novel geospatial deception metric; submitted to the NeurIPS 2026 Workshop: Tackling Climate Change with Machine Learning.
- Extending the study toward a journal submission: benchmarking additional geospatial VLMs (GeoChat, GeoPixel) against the deception metric, exploring prompt-engineering and image-transformation strategies (boundary overlays, supplementary maps) to improve grounding, and developing a sequential VLM pipeline to extract and collate temporal information for tracking evolving events such as hurricanes

PUBLICATIONS

P. S. Nishant, S. A. Nistala, U. V. Rohith, and S. Upadhyaya, "Spatio-Temporal Reconstruction of MODIS Land Surface Temperature over Hyderabad Using Convolutional LSTM," *IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2026)*.

P. S. Nishant, K. H. Singh, M. Duddi, S. Mukherjee, and B. Bhattacharyya, "Bayesian Physics-Informed Neural Network for Elastic Modulus and Poisson's Ratio Estimation of Concrete via Embedded PZT Sensor," *Structural Engineering Convention (SEC 2026)*, IIT Hyderabad, Dec. 20-23, 2026.

POSITIONS OF RESPONSIBILITY

General Secretary | Civil Engineering Society (CES), IIT Hyderabad

- Coordinated departmental seminars, orientations, and technical events involving 100+ participants, managing logistics, hospitality, and inter-team operations.
- Facilitated student engagement initiatives and collaborated with faculty, speakers, and vendors for smooth execution of academic and professional events.

SKILLS

Software/Tools: AutoCAD, QGIS, SIDRA Intersection, Python (PyTorch, NumPyro, Scikit-learn, NumPy, Pandas), Microsoft Office

Core Areas: Transportation Engineering, Traffic Analysis, Physics-Informed Machine Learning, Uncertainty Quantification

RELEVANT COURSES

Institute Coursework: Machine Learning and Scientific Computing, Mathematical Methods, Traffic Engineering and Planning, Highway Design, Probability & Statistics

Certifications: Supervised Machine Learning: Regression and Classification (Andrew Ng), Advanced Learning Algorithms (Andrew Ng), AutoCAD 2D & 3D (Mike Freeman)

EXTRACURRICULAR

Member, IIT Hyderabad Basketball Team -- represented the institute in local tournaments.